

Effects of Caffeine Ingestion on Exercise Performance at Low and High Altitudes in Cross-Country Skiers

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Abstract

B. Berglund and P. Hemmingsson, Effects of Caffeine Ingestion on Exercise Performance at Low and High Altitudes in Cross-Country Skiers. Int J Sports Med, Vol 3, No 4, pp 234–236, 1982.

The effect of caffeine on exercise time over a specified distance (approx. 21 km) was investigated in well-trained cross-country skiers (14 subjects) both at low (300 m above sea level) and high (2900 m above sea level) altitudes. Each subject participated in two races at both altitudes – one after caffeine ingestion (6 mg/kg body weight) and one after placebo on a double-blind basis. In each race 50 % of the participants were medicated with caffeine, thereby after normalization (where the mean time was set at 100 %), when the influence of weather and snow conditions were minimized, the difference between the race time with caffeine and placebo could be calculated for each individual and expressed as a percentage of the mean.

At low altitudes, after half the distance, the race time with caffeine was decreased 0.94 % of the mean time (corresponding to approx. 33 s) as compared to placebo ($P < 0.05$), whereas after the full distance the corresponding decrease was 1.7 % of the mean time (corresponding to approx. 59 s) ($0.05 < P < 0.10$). At high altitudes, the race time was significantly faster with caffeine than with placebo ($P < 0.001$), both after one and two laps, 2.19 % and 3.18 % of the mean time, respectively (corresponding to approx. 101 s and 152 s). The perceived exertion was similar at low and high altitudes independent of medication.

In conclusion, there are indications that caffeine induces an increased performance capacity for cross-country skiers at low altitudes. Furthermore, an even more marked improvement after caffeine as compared to placebo was found at high altitudes.

Key words: caffeine, exercise performance, hypoxia, high altitude, perceived exertion

Introduction

Recent observations suggest that physical work at a standardized submaximal load can be maintained significantly longer after the administration of caffeine (2). This effect was suggested to be due to a glycogen sparing effect in combination with a lessening of the subjective perception of effort. Furthermore, caffeine seems to increase work production during long-term isokinetic cycling exercise (6). It has, however, not yet been fully evaluated whether this also means that a certain racing distance can be fulfilled in a shorter time. Therefore, in the present study we examined the effect of caffeine on the exercise time over a

specified distance in cross-country skiing both at low and high altitudes.

Material and Methods

The first set of tests was performed at 2900 m above sea level and began after 3 days of acclimatization. The participants were five female and eight male healthy competitive cross-country skiers. These subjects were 20–30 years of age and were very well-trained with a training time of 430–700 h per year. The skiers consumed the same type of mixed diet and performed the same type of training program on the 3 days preceding as well as throughout the investigation. On separate days, four races were performed, two with females and two with males. Each subject participated in two races approx. 2 h after a light meal and the races were preceded by a caffeine-free period of at least 12 h. One hour prior to the start each subject was given a gelatin capsule containing either caffeine (6 mg/kg body weight) or placebo on a double-blind basis. The dose administered corresponds to approximately three cups of coffee. The caffeine dose was in general well tolerated. In the first set of tests, two subjects who received caffeine and one who received the placebo complained of nausea, whereas in the second set of tests three subjects receiving the active drug and one the placebo had the same symptom. When the double-blind code was broken, it was demonstrated that approximately 80 % of the subjects could distinguish between caffeine and placebo. However, there was no difference in this respect at low and high altitudes. The time and perceived exertion ad modum Borg (1) was established after one (approx. 11.5 km) and two laps (approx. 23 km).

The second set of tests was performed approx. 300 m above sea level about 5 weeks after the first set of tests and after an acclimatization period of approximately 4 weeks. Four female and ten male well-trained healthy competitive cross-country skiers of similar ages and similarly well-trained participated in the second set of tests. Similar preparations were made as for the first tests with the exception that the racing course was slightly shorter, about 20 km.

Normalization Procedure

To minimize the influence of weather and snow conditions, a normalization procedure was performed. In each set of tests, the skiers performed one race after caffeine medication and one after placebo medication, and in both races

50% of the participants were medicated with caffeine and 50% with placebo. It was then assumed that the influence of the active drug was similar in each race and, therefore, the group mean time after the full distance in each race was set at 100%. For each individual, the race time in the first and second race was taken as the percent of group mean in the first and second race, respectively. The difference in race time (in percent of group mean time) between the race after caffeine and placebo medication was then calculated for each skier.

In the first race, an uneven number of subjects participated. Therefore, one subject, by chance, was eliminated from the normalization procedure so that 50% were medicated with caffeine when the normalization procedure was performed. After this normalization procedure, the race time of the "uneven" subject was expressed in percent of group mean of those participating in the normalization procedure in the first and second race, respectively. Thereafter, the difference between the race after caffeine and placebo medication was calculated as above for that subject.

All participants were volunteers and gave their written consent. The study was approved by the Ethical Committee at the Karolinska Hospital. Student's *t* test was used for comparison of groups.

Results

At Low Altitude

In females, the mean time was 67 min 10 s in the first and 67 min 20 s in the second race, whereas in males the corresponding values were 55 min 33 s and 55 min 25 s, respectively.

When comparing the races with caffeine and placebo, it was found that all females and seven of the ten males completed the first lap in a shorter time after caffeine medication than after placebo medication (expressed as percent of mean time), whereas the same applied to all females and six of ten males over two laps.

When males and females were combined, the paired *t* test revealed a significant difference ($P < 0.05$) between the race after caffeine and placebo medication over one lap. The corresponding difference ($1.70 \pm 3.10\%$) over two laps was not statistically significant ($t = 2.06$; $0.05 < P < 0.10$) (Table 1).

The difference in race time (expressed as percent of mean time) between caffeine and placebo medication over one lap corresponded to approx. 38 s in females and 31 s in males. Over two laps, the corresponding values were 65 s and 56 s, respectively. The perceived exertion was similar, independent of medication, both after one and two laps (Table 1).

At High Altitude

Due to extremely different weather and snow conditions, the mean times between races varied fairly much. Thus, in females the mean time of those four participating in the normalization procedure was 96 min 48 s in the first and 84 min 42 s in the second race. The corresponding mean times in eight males were 68 min 25 s and 73 min 44 s, respectively.

When comparing the race with caffeine to the race with placebo, it was found that when medicating with caffeine four of five females and all males were faster (expressed as percent of mean time) both over one and two laps than with placebo. Consequently, the paired *t* test revealed a significant difference ($P < 0.001$) between the race with placebo and caffeine both after one and two laps (Table 1). After the first lap, this difference corresponded to approx. 119 s in females and 93 s in males. After two laps, the corresponding values were approx. 174 s and 142 s, respectively. On the other hand, the perceived exertion after both one and two laps was essentially the same with caffeine and placebo (Table 1).

Discussion

In the present study, there are indications that caffeine induces an increased performance capacity for cross-country ski-running at low altitudes. This finding is in accordance with previous data by Costill and co-workers (2, 6). In addition, at high altitudes there was an even more marked improvement of performance capacity after caffeine as compared to placebo medication.

Some factors may, alone or in combination, have caused the increased physical performance after caffeine medication. On the one hand, it has been suggested that caffeine has a glycogen sparing effect possibly by accelerating lipolysis, thereby making more free fatty acids available earlier during physical exercise. This is supported by the finding

Tab. 1 Time (in percent of mean in either first or second race) and perceived exertion (PE) ad modum Borg after medication with placebo or caffeine at low and high altitudes

Site	No	Male/ Female	Placebo (P)				Caffeine (C)				Difference P-C			
			1 lap Time of % mean	PE	2 laps Time of % mean	PE	1 lap Time of % mean	PE	2 laps Time of % mean	PE	1 lap Time of % mean	PE	2 laps Time of % mean	PE
At low altitude (300 m)	14	10/4	49.77 ±2.08	14.64 ±1.41	100.85 ± 3.82	14.86 ±1.23	48.83 ±1.51	14.14 ±0.93	99.15 ±3.42	14.46 ±1.69	0.94* ±1.45	0.50 ±1.4	1.70 ±3.10	0.40 ±1.80
At high altitude (2900 m)	13	8/5	51.35 ±1.84	13.81 ±1.18	102.17 ± 3.81	14.69 ±1.38	49.16 ±2.39	14.35 ±1.52	98.99 ±3.93	14.69 ±1.25	2.19*** ±1.70	0.54 ±2.10	3.18*** ±2.18	0.00 ±1.68

*, *** indicates a significance of $P < 0.05$ and $P < 0.001$, respectively.

that the work load at which the lactate accumulation in blood starts is increased (7). On the other hand, the stimulatory effect of caffeine on the nervous system is well established (10) and must be taken into account when considering any improvements in physical performance after administration of this drug.

Since the working times were fairly short, it seems unlikely that the glycogen stores have been completely emptied. Therefore, the delay of glycogen depletion does not seem to explain the improved performance capacity. This idea is supported by the observation of an increased speed also after the first lap.

Farrell et al. (3) suggested that distance runners appear to set a race pace that allows the largest possible $\dot{V}O_2$, which just avoids the exponential rise in plasma lactate (the lactate threshold). Due to the increased availability of free fatty acids (FFA), caffeine seems to increase this lactate threshold (7). Therefore, it seems reasonable to believe that by this mechanism the increased FFA levels can contribute to the higher ski-running speed after caffeine as compared to placebo medication.

During ergometer bicycling at a specified work load, the exercise time to exhaustion was significantly increased after caffeine ingestion (2). However, during this exercise period after caffeine, the perceived exertion was also lower as compared to placebo medication. In the present study, the skiers chose a similar level of exertion, both after one and two laps, independent of medication. Consequently, it seems reasonable that the skiers increased their speed to obtain the same level of perceived exertion after caffeine as after placebo medication. This may reflect the stimulatory effect on the nervous system of caffeine but may also reflect a difference in fatigue-producing metabolic products of work.

As stated above, a metabolic and a nervous stimulatory mechanism either alone or in combination seem to be beneficial for long-term performance capacity after caffeine ingestion. However, the present study cannot differentiate between those two.

Although the performances at high altitudes were affected to some extent by the varying weather conditions, our data suggest a more pronounced improvement of performance capacity at high altitudes after caffeine ingestion. The reason for this effect is not known.

Keul et al. (9) reported higher plasma lactate levels, an increased glucose metabolism, and decreased FFA metabolism during "maximal steady state work" at 4000 m as compared to sea level. These observations also seem to be valid at approx. 2500 m where Kaijser (8) found increased plasma lactate levels and shorter working time to exhaustion as compared to sea level. These data may suggest that

the lactate threshold is decreased and therefore that, possibly, the range by which this threshold can be influenced by increased FFA levels is widened. However, these data were not confirmed by Hermansen et al. (5) who demonstrated similar plasma lactate levels after exercise performed at equal percentage of actual $\dot{V}O_{2\max}$ at high and low altitudes.

During hypoxia, an increased sensitivity for some drugs has been reported (4), but whether this also applies for caffeine is not known.

Another factor to consider in this respect is the time between the sets of tests. Although there was only a difference of 1 month, it cannot be fully excluded that the skiers were less trained during the first set of tests (at high altitude) as compared to the second one and that the beneficial effect of caffeine was lower after physical training.

Acknowledgment

The authors are grateful to the skiers for stimulating collaboration and for having changed their training schedule to make this study possible. Åke Jönsson and Jan Halvarsson are thanked for valuable advice and excellent assistance. We are also grateful to Professor Lennart Kaijser for valuable discussions and for having read the manuscript.

This study was supported by a grant from the Swedish Sports Research Council (No. 25/81).

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